Chrome File Edit View History Bookmarks Profiles Tab Window Help Fri 2 Aug 6:25 AM

HackerRank Question

hackerrank.com/test/g1lj816k4tp/questions/apm6j4dps8d

1h 14m left

ALL

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1. Quiz Competition

A class has a number of students, each with a particular area of knowledge, or *talent*. Each talent is represented as an integer from *1* to *talentsCount*. Teams must be formed for a quiz competition, and must have at least one member with each of the talents. The talents are listed in an array, and the students must be chosen consecutively, starting at any element index. For each starting index, determine the minimum number of students that must be selected to have at least one team member with each talent. If it is not possible, return -1 for that index.

Example

talentsCount = 3

talent = [1, 2, 3, 2, 1]

There is no way fewer than *3* students can have at least *3* talents. The following subarrays of length *3* or more are possible:

Starting at position 1: ***[1, 2, 3]***, *[1, 2, 3, 2]*, *[1, 2, 3, 2, 1]*

Starting at position 2: *[2, 3, 2]*, ***[2, 3, 2, 1]***

Starting at position 3: ***[3, 2, 1]***

No further subarrays of length 3 or more exist.

Language C++20 Environment Autocomplete Ready

```cpp
#include <bits/stdc++.h>...

/*
 * Complete the 'teamSize' function below.
 *
 * The function is expected to return an INTEGER_ARRAY.
 * The function accepts following parameters:
 *  1. INTEGER_ARRAY talent
 *  2. INTEGER talentsCount
 */

vector<int> teamSize(vector<int> talent, int talentsCount) {

}

int main()...
```

Line: 9 Col: 1

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HackerRank Question -

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1h 12m left

ALL

1 3 4 5 6 7 8

talent = *[1, 2, 3, 2, 1]*

There is no way fewer than *3* students can have at least *3* talents. The following subarrays of length *3* or more are possible:
Starting at position 1: ***[1, 2, 3]***, *[1, 2, 3, 2]*, *[1, 2, 3, 2, 1]*
Starting at position 2: *[2, 3, 2]*, ***[2, 3, 2, 1]***
Starting at position 3: ***[3, 2, 1]***
No further subarrays of length 3 or more exist.

Starting at the first position, the shortest subarray with one of each talent value has length 3.
Starting at element 2, the shortest has length 4.
At position 3, the shortest and only possible subarray has length 3.
It is not possible to form a team with fewer than 3 members, so the subarrays starting at the last two positions will fail.

The highlighted subarrays are selected. The return array is *[3, 4, 3, -1, -1]*.

Function Description

Complete the function *teamSize* in the editor below. It must return an array of integers.

teamSize has the following parameter(s):
talent: An array of length *n* where *talent[i]*

Language C++20 Environment Autocomplete Ready

```
#include <bits/stdc++.h>···

/*
 * Complete the 'teamSize' function below.
 *
 * The function is expected to return an INTEGER_ARRAY.
 * The function accepts following parameters:
 *  1. INTEGER_ARRAY talent
 *  2. INTEGER talentsCount
 */

vector<int> teamSize(vector<int> talent, int talentsCount) {

}

int main()···
```

Line: 9 Col: 1

Test Results Custom Input Run Code Run Tests Submit

1h 12m left

ALL

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5

6

7

8

return array is [3, 4, 3, -1, -1].

Function Description

Complete the function *teamSize* in the editor below. It must return an array of integers.

teamSize has the following parameter(s):

talent: An array of length *n* where *talent[i]* denotes the talent of the i^{th} student, where $0 \le i < n$.

talentsCount: An integer, the total number of talents represented

Constraints

- $1 \le n, talentsCount \le 10^5$
- $1 \le talent[i] \le talentsCount$

▶ **Input Format For Custom Testing**

▼ **Sample Case 0**

Sample Input For Custom Testing

```
8
1
1
2
2
```

Language C++20 Environment Autocomplete Ready

```
#include <bits/stdc++.h>

/*
 * Complete the 'teamSize' function below.
 *
 * The function is expected to return an INTEGER_ARRAY.
 * The function accepts following parameters:
 *  1. INTEGER_ARRAY talent
 *  2. INTEGER talentsCount
 */

vector<int> teamSize(vector<int> talent, int talentsCount) {

}

int main()
```

Line: 9 Col: 1

Test Results Custom Input Run Code Run Tests Submit

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ALL

- *1 ≤ talent[i] ≤ talentsCount*

▸ **Input Format For Custom Testing**

▾ **Sample Case 0**

Sample Input For Custom Testing

```
8
1
1
2
2
3
1
3
2
3
```

Sample Output

```
5
4
4
3
4
3
-1
-1
```

Explanation

n = 8

talent = [1, 1, 2, 2, 3, 1, 3, 2]

talentsCount = 3

Language C++20 Environment Autocomplete Ready

```cpp
#include <bits/stdc++.h>

/*
 * Complete the 'teamSize' function below.
 *
 * The function is expected to return an INTEGER_ARRAY.
 * The function accepts following parameters:
 *  1. INTEGER_ARRAY talent
 *  2. INTEGER talentsCount
 */

vector<int> teamSize(vector<int> talent, int talentsCount) {

}

int main()
```

Line: 9 Col: 1

Test Results Custom Input Run Code Run Tests Submit

1h 12m left

ALL

1 3 4 5 6 7 8

```
3
-1
-1
```

Explanation

n = 8

talent = [1, 1, 2, 2, 3, 1, 3, 2]

talentsCount = 3

There have to be at least *3* elements in any subarray to represent each talent. The following subarrays are possible:

Beginning at the first position: [1, 1, 2], [1, 1, 2, 2], **[1, 1, 2, 2, 3]**, [1, 1, 2, 2, 3, 1], [1, 1, 2, 2, 3, 1, 3], [1, 1, 2, 2, 3, 1, 3, 2]

Second position: [1, 2, 2], **[1, 2, 2, 3]**, [1, 2, 2, 3, 1], [1, 2, 2, 3, 1, 3], [1, 2, 2, 3, 1, 3, 2]

Third position: [2, 2, 3], **[2, 2, 3, 1]**, [2, 2, 3, 1, 3], [2, 2, 3, 1, 3, 2]

Fourth position: **[2, 3, 1]**, [2, 3, 1, 3], [2, 3, 1, 3, 2]

Fifth position: [3, 1, 3], **[3, 1, 3, 2]**

Sixth position: **[1, 3, 2]**

No further subarrays will have 3 elements.

The highlighted subarrays are the shortest for each starting position. The return array of subarray lengths is *[5, 4, 4, 3, 4, 3, -1, -1]*

▸ **Sample Case 1**

▸ **Sample Case 2**

Language C++20 Environment Autocomplete Ready

```cpp
#include <bits/stdc++.h>

/*
 * Complete the 'teamSize' function below.
 *
 * The function is expected to return an INTEGER_ARRAY.
 * The function accepts following parameters:
 *  1. INTEGER_ARRAY talent
 *  2. INTEGER talentsCount
 */

vector<int> teamSize(vector<int> talent, int talentsCount) {

}

int main()
```

Line: 9 Col: 1

Test Results Custom Input Run Code Run Tests Submit

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1h 12m left

ALL

1

3

4

5

6

7

8

The highlighted subarrays are the shortest for each starting position. The return array of subarray lengths is *[5, 4, 4, 3, 4, 3, -1, -1]*

▼ Sample Case 1

Sample Input For Custom Testing

```
5
1
1
1
1
1
1
```

Sample Output

```
1
1
1
1
1
```

Explanation

n = 5

talent = [1, 1, 1, 1, 1]

talentsCount = 1

Since there is only *1* type of talent, any *1* person can form a team.

▶ Sample Case 2

Language C++20   Environment   Autocomplete Ready

```
#include <bits/stdc++.h>

/*
 * Complete the 'teamSize' function below.
 *
 * The function is expected to return an INTEGER_ARRAY.
 * The function accepts following parameters:
 *  1. INTEGER_ARRAY talent
 *  2. INTEGER talentsCount
 */

vector<int> teamSize(vector<int> talent, int talentsCount) {

}

int main()
```

Line: 9 Col: 1

Test Results   Custom Input   Run Code   Run Tests   Submit

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HackerRank Question -

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1h 12m left

ALL

1

3

4

5

6

7

8

Since there is only *1* type of talent, any *1* person can form a team.

▼ Sample Case 2

Sample Input For Custom Testing

```
9
7
5
3
4
6
1
7
2
4
7
```

Sample Output

```
8
7
-1
-1
-1
-1
-1
-1
-1
```

Explanation

n = 9

talent = [7, 5, 3, 4, 6, 1, 7, 2, 4]

talentsCount = 7

Language C++20 Environment Autocomplete Ready

```
#include <bits/stdc++.h>

/*
 * Complete the 'teamSize' function below.
 *
 * The function is expected to return an INTEGER_ARRAY.
 * The function accepts following parameters:
 *  1. INTEGER_ARRAY talent
 *  2. INTEGER talentsCount
 */

vector<int> teamSize(vector<int> talent, int talentsCount) {

}

int main()
```

Line: 9 Col: 1

Test Results Custom Input Run Code Run Tests Submit

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HackerRank Question

hackerrank.com/test/g1lj816k4tp/questions/apm6j4dps8d

1h 12m left

ALL

1

3

4

5

6

7

8

```
4
7
```

Sample Output

```
8
7
-1
-1
-1
-1
-1
-1
-1
```

Explanation

n = 9

talent = [7, 5, 3, 4, 6, 1, 7, 2, 4]

talentsCount = 7

There have to be at least *7* elements in any subarray to represent each talent. The following subarrays are possible:

Beginning at the first position: *[7, 5, 3, 4, 6, 1, 7]*, ***[7, 5, 3, 4, 6, 1, 7, 2]***, *[7, 5, 3, 4, 6, 1, 7, 2, 4]*

Second position: ***[5, 3, 4, 6, 1, 7, 2]***, *[5, 3, 4, 6, 1, 7, 2, 4]*

Third position: *[3, 4, 6, 1, 7, 2, 4]*

No further subarrays will have 7 elements.

The highlighted subarrays are the shortest for each starting position. The return array of subarray lengths is *[8, 7, -1, -1, -1, -1, -1, -1, -1]*.

Language C++20 Environment Autocomplete Ready

```cpp
#include <bits/stdc++.h>

/*
 * Complete the 'teamSize' function below.
 *
 * The function is expected to return an INTEGER_ARRAY.
 * The function accepts following parameters:
 *  1. INTEGER_ARRAY talent
 *  2. INTEGER talentsCount
 */

vector<int> teamSize(vector<int> talent, int talentsCount) {

}

int main()
```

Line: 9 Col: 1

Test Results Custom Input Run Code Run Tests Submit

ALL

Determine if a triangle formed by three points a(x1, y1), b(x2, y2), and c(x3, y3) is non-degenerate. If two given points, p = (xp, yp) and q = (xq, yq), are located on or inside a triangle, they *belong* to the triangle. Return the correct scenario number.

Scenarios:

- *0*: the lines do not form a valid non-degenerate triangle
- *1*: point *p* belongs to the triangle but point *q* does not
- *2*: point *q* belongs to the triangle but point *p* does not
- *3*: both points *p* and *q* belong to the triangle
- *4*: neither point *p* nor point *q* belong to the triangle

Note: A triangle is considered non-degenerate if it meets the following conditions, where |ab| denotes the length of the line segment between points a and b:

- |ab| + |bc| > |ac|
- |bc| + |ac| > |ab|
- |ab| + |ac| > |bc|

Example

```
1 =
a(x1,y1):
(2,2)
2 =
b(x2,y2):
```

```
double area(int x1, int y1, int x2, int y2, int x3, int y3) {
 return fabs(x1*(y2-y3) + x2*(y3-y1) + x3*(y1-y2)) / 2.0;
}
```

Search

ALL

1

2

3

4

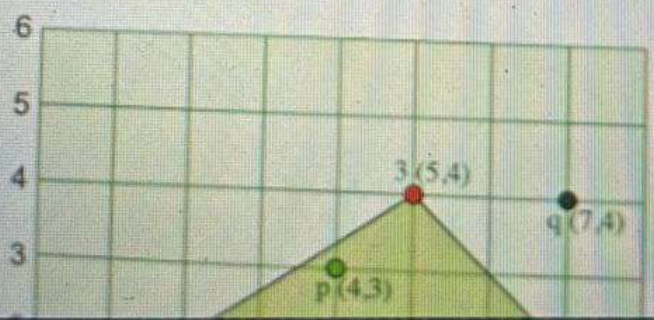
5

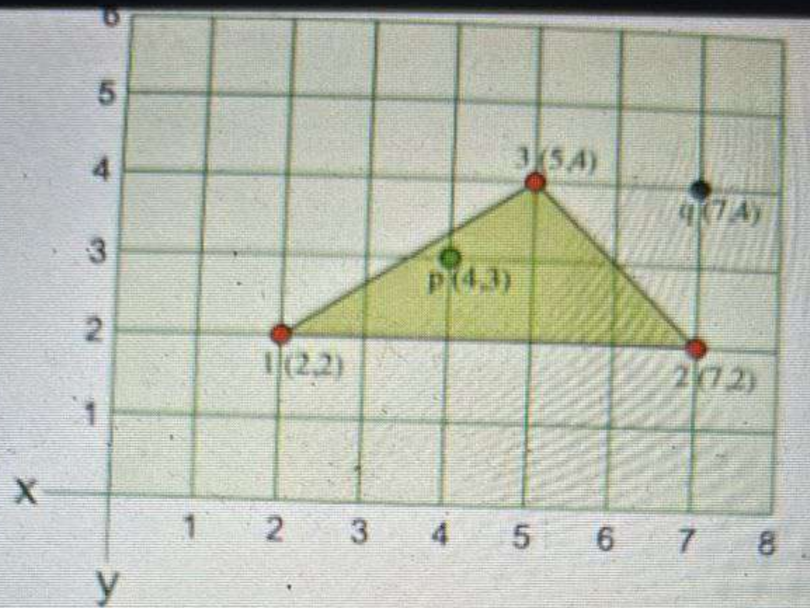
Note: A triangle is considered non-degenerate if it meets the following conditions, where |ab| denotes the length of the line segment between points a and b:

- |ab| + |bc| > |ac|
- |bc| + |ac| > |ab|
- |ab| + |ac| > |bc|

Example

```
    1 =
a(x1,y1):
(2,2)
    2 =
b(x2,y2):
(7,2)
    3 =
c(x3,y3):
(5,4)
    p = p(xp,
yp): (4,3)
    q = q(xq,
yq): (7,4)
```





First, the triangle abc forms a valid non-degenerate triangle.

- $|ab| = 7 - 2 = 5$. $|bc| = sqrt((7-5)^2 + (4-2)^2) = sqrt(2^2 + 2^2) = sqrt(8) = 2.82$. $|ac| = sqrt((5 - 2)^2 + (4 - 2)^2) = (3^2 + 2^2) = sqrt(13) = 3.6$.
- $|ab| + |bc| > |ac|$ => $5 + 2.82 > 3.6$
- $|bc| + |ac| > |ab|$ => $2.82 + 3.6 > 5$
- $|ab| + |ac| > |bc|$ => $5 + 3.6 > 2.82$

Second, the point p(5, 4) belongs to the triangle but point q(7, 4) does not as shown. So, the answer is 1.

Function Description

Complete the function *pointsBelong* in the editor below.

pointsBelong has the following parameter(s):

Search

- |ab| = 7 - 2 = 5. |bc| = sqrt((7-5)^2 + (4-2)^2) = sqrt(2^2 + 2^2) = sqrt(8) = 2.82. |ac| = sqrt((5 - 2)^2 + (4 - 2)^2) = (3^2 + 2^2) = sqrt(13) = 3.6.
- |ab| + |bc| > |ac| => 5 + 2.82 > 3.6
- |bc| + |ac| > |ab| => 2.82 + 3.6 > 5
- |ab| + |ac| > |bc| => 5 + 3.6 > 2.82

Second, the point p(5, 4) belongs to the triangle but point q(7, 4) does not as shown. So, the answer is 1.

Function Description

Complete the function *pointsBelong* in the editor below.

pointsBelong has the following parameter(s):

int x1, y1, x2, y2, x3, y3: integer coordinates of the three points that may create a valid triangle

int xp, yp, xq, yq: integer coordinates of the two points *p* and *q*

Returns

int: the scenario number

Constraints

- $0 \le x1, y1, x2, y2, x3, y3, xp, yp, xq, yq \le 2000$

▸ Input Format for Custom Testing

▾ Sample Case 0

Search

Returns

int: the scenario number

Constraints

- $0 \le x1, y1, x2, y2, x3, y3, xp, yp, xq, yq \le 2000$

▶ Input Format for Custom Testing

▼ Sample Case 0

Sample Input 0

```
STDIN     Function
-----     --------
0    →    (x1,y1) = (0,0)
0
2    →    (x2,y2) = (2,0)
0
4    →    (x3,y3) = (4,0)
0
2    →    p = (xp,yp) = (2,0)
0
4    →    q = (xq,yq)=(4,0)
0
```

Sample Output 0

```
0
```

Explanation 0

kerRank Question - Wi

https://www.hackerrank.com/test/g1lj816k4tp/questions/ldcs7skgj

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ACLE FY'26 - Campus Hiring - IIT Guwahati - Internship_2 Aug Answer

2. Why is a virtual memory address called 'virtual'?

Why is a virtual memory address called 'virtual'?

Pick **ONE** option

- [✓ This memory address does not map to actual memory cell that corresponds to address number.]
- This memory is the same as physical memory.
- This memory cannot be accessed by applications at run time.
- This memory address is always present even after a system is restarted.

Solve question 3

Clear Selection

5

6

7

8

9

3. Consider the two functions incr and decr shown below. incr(){ decr(){ wait(s
are 5 threads each invoking incr once, and 3 threads each invoking decr once
shared variable X is 10 Suppose there are two implementations of the semap
initialized to 1. I-2: S is a counting semaphore initialized to 2. Let V1 and V2 b

https://www.hackerrank.com/test/g1lj816k4tp/questions/gdc5p221dlq

ORACLE FY'26 - Campus Hiring - IIT Guwahati - Internship_2 Aug

ALL

1
2
3
4
5
6
7
8
9

4. What is the main difference between "user mode" and "kernel mode"

What is the main difference between "user mode" and "kernel mode" in an operating system?

Pick **ONE** option

- User mode allows direct hardware access, kernel mode does not.
- Kernel mode can access protected memory regions, user mode cannot.
- Kernel mode applications are given less priority than user mode applications.
- User mode instructions can switch the CPU to kernel mode, but not vice-versa.

Clear Selection

5. Which data structure is best suited for implementing a priority q

Which data structure is best suited for implementing a priority queue?

https://www.hackerrank.com/test/g1lj816k4tp/questions/c3aaf7oq0e6

ALL

1
2
3
4
5
6
7
8
9

5. Which data structure is best suited for imp

Which data structure is best suited for implementing a priority q

Pick **ONE** option

- Array
- Slack
- Heap
- Linked List

Clear Selection

6. In the context of Dynamic Programming,

In the context of Dynamic Programming, what is "memoizatio

https://www.hackerrank.com/test/g1lj816k4tp/questions/a7dn8s3ffm6

In the context of Dynamic Programming, what is "memoization"?

In the context of Dynamic Programming, what is "memoization"?

Pick **ONE** option

- Writing an algorithm in a memo to remember it
- Recomputing solutions to subproblems multiple times for accuracy
- Storing solutions to subproblems to avoid recomputation
- Optimizing a problem by breaking it into subproblems and solving iteratively

Clear Selection

7. A certain program must store the names of all the people who finish the finish line as the competitors finish. After the race is over the prog competitors who placed 1 to 10 and then 100, 200, 300 and so on, as the structure would be the best choice for this program?

printf("%c\n", *p); return 0; }

```
#include <stdio.h>
int main() {
    char arr[] = "abcd";
    char *p = arr;
    printf("%c\t", ++*p);
    printf("%c\t", *p++);
    printf("%c\t", (*p)++);
    printf("%c\n", *p);
    return 0;
}
```

6

8

9

10

11

12

Pick **ONE** option

- b b b c ✓
- b c c d
- b b c c
- b c c c

Clear Selection

https://www.hackerrank.com/test/g1lj816k4tp/questions/3g83a54sl0o

ORACLE

ALL

6

8

10

12

14

10. What is the latest long-term release of Oracle Database?

What is the latest long-term release of Oracle Database?

Pick **ONE** option

- 23ai
- 22ai
- 24ai
- 19ai

Clear Selection

11. Which of the following is not a property of

Which of the following is not a property of Relational Database

https://www.hackerrank.com/test/g1lj816k4tp/questions/1qed2r1nbfj

2. Which SQL constraint do we use to set some value to a field whose value ...

Which SQL constraint do we use to set some value to a field whose value ...

Pick **ONE** option

- NOT NULL
- DEFAULT ✓
- CHECK
- UNIQUE

Clear Selection

13. Truncate table is classified as:

Truncate table is classified as:

www.hackerrank.com/test/g1lj816k4tp/questions/1pdsi1nm44h

14. How will you distribute incoming traffic to a set of Web Servers?

How will you distribute incoming traffic to a set of Web Servers?

Pick **ONE** option

- By deploying a Firewall before the servers
- By deploying a router before the servers.
- By deploying a load balancer before the servers.
- By deploying a switch before the servers.

Clear Selection

15. In a distributed system, what is CAP theorem?

In a distributed system, what is CAP theorem?

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15. In a distributed system, what is CAP theorem?

In a distributed system, what is CAP theorem?

Pick **ONE** option

- It specifies the maximum allowable response time for a system
- It describes the trade-offs between consistency, availability, and partition tole
- It defines the minimum number of servers required for fault tolerance.
- It outlines the process for scaling a system horizontally.

Clear Selection

18

16. What is a Zombie process in Linux?

19

What is a Zombie process in Linux?

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What is a Zombie process in Linux ?

t is a Zombie process in Linux ?

ONE option

- A process which is running in the background.
- A process which is running for a long time.
- A process whose execution has completed but it still has an entry in the process table.
- A process which has crashed due to memory corruption.

Clear Selection

Saved!

17. Which HTTP method is suitable for resource creation?

Which HTTP method is suitable for resource creation?

Pick ONE option

Which HTTP method is suitable for resource creation?

ch HTTP method is suitable for resource creation?

k **ONE** option

- PUT
- PATCH
- POST
- CREATE

Clear Selection

18. Which response status code category represe

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29. What is the remainder when 20001000 divided by 13?

What is the remainder when 2000^{1000} divided by 13?

Pick **ONE** option

- 3
- 4
- 8
- 11

Clear Selection

30

31

32

33

30. Interpret Charts

The following graph shows the number of people (in millions) using Hinge... the market

https://www.hackerrank.com/test/g1lj816k4tp/questions/75eqgqdki3o

1. Rearrange Students

In your school, two lines of students are formed A and B, with each line having exactly N students. The two lines of students will face each other. The Physical Education teacher wants the heights of all students standing across from each other to be equal. To do this, students may be swapped from one line to another. Any student in one line can be swapped with any student in the other line. Each swap counts as one operation, and each operation costs the minimum height in the swap. Reordering the students within one line has no cost. Determine the minimal cost to achieve the desired result. If it is impossible, return -1.

Example

arrA = [4, 2, 2, 2]

arrB = [1, 4, 1, 2]

If the arrays are sorted descending, they are arrA = [4, 2, 2, 2] and arrB = [4, 2, 1, 1]. Swap a 2 in arrA with a 1 in arrB for a cost of min(1, 2) = 1. This is the only swap that needs to occur so the answer is 1.

Function Description

Complete the function *rearrangeStudents* in the editor below.

rearrangeStudents has the following parameters:

int arrA[n]: the heights of children in line A

int arrB[n]: the heights of children in line B

In your school, two lines of students are formed A and B, with each line having exactly N students. The two lines of students will face each other. The Physical Education teacher wants the heights of all students standing across from each other to be equal. To do this, students may be swapped from one line to another. Any student in one line can be swapped with any student in the other line. Each swap counts as one operation, and each operation costs the minimum height in the swap. Reordering the students within one line has no cost. Determine the minimal cost to achieve the desired result. If it is impossible, return -1.

Example

arrA = [4, 2, 2, 2]

arrB = [1, 4, 1, 2]

If the arrays are sorted descending, they are *arrA* =[4, 2, 2, 2] and *arrB* = [4, 2, 1, 1]. Swap a *2* in *arrA* with a *1* in *arrB* for a cost of *min(1, 2)* = *1*. This is the only swap that needs to occur so the answer is *1*.

Function Description

Complete the function *rearrangeStudents* in the editor below.

rearrangeStudents has the following parameter

int *arrA[n]*: the heights of children in line A

int *arrB[n]*: the heights of children in line B